

**Transcriptomic Analysis of Respiratory Epithelial Cellular Responses Following Cigarette
Smoke Exposures**

Meena Easwaran

McWilliams School of Biomedical Informatics
The University of Texas Health Science Center at Houston

BMI 5332 Statistical Analysis of Genomic Data

Dr. Arif Harmanci

Spring 2024

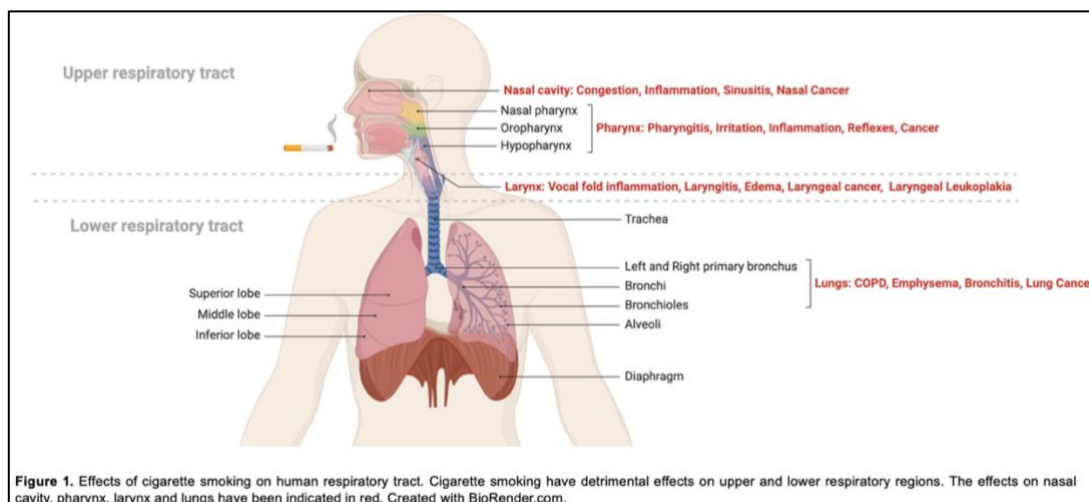
April 25, 2024

Problem Statement

How do human and mouse transcriptomic responses differ and overlap following exposure to cigarette smoke in non-laryngeal airway regions within air-liquid Interface cell cultures?

Significance and Background

Cigarette smoking adversely impacts respiratory health in several parts of the respiratory system, such as the nasal passages (Hadar et al., 2009; Pagliuca et al., 2015), pharynx (Garcia Martins et al., 2012; Moore, 1971; Radsel & Kambe, 1978), larynx (Easwaran et al., 2021, 2022; Erickson-DiRenzo et al., 2020, 2021; Moore, 1971; R. A. Renne & Gideon, 2006), and lungs (Curtin et al., 2004; Phillips et al., 2015; R. Renne et al., 2009; Walser et al., 2008). A variety of pathological conditions (**Figure 1**) can be initiated or aggravated by the inhalation of cigarette smoke (CS) due to its high concentration of toxicants, including carcinogens (Centers for Disease Control and Prevention, 2010; Services et al., 2014). CS exposure causes chronic rhinosinusitis and nasal congestion in the nasal passageways, the first organs affected within the respiratory or airway systems (Reh et al., 2012). Moreover, chronic CS-induced irritation and inflammation of the nasal mucosa can also increase the risk of developing nasal tumors (Vineis et al., 2004).



Prolonged exposures to CS can also affect the pharynx and larynx, resulting in conditions such as pharyngitis, laryngitis, Reinke's edema, and cancers (Easwaran et al., 2021, 2022; Erickson-DiRenzo et al., 2020, 2021; Garcia Martins et al., 2012; Moore, 1971; Radsel & Kambe, 1978; R. A. Renne & Gideon, 2006).

With the progression downwards towards the lower airways, particularly the lungs, the negative effects of CS become more prominent. Chronic obstructive pulmonary disease (COPD) is a major cause of global mortality, often caused by prolonged exposure to CS (Laniado-Laborin, 2009; Song et al., 2021). Patients with Chronic Obstructive Pulmonary Disease (COPD), including those who smoke, often show symptoms such as chronic bronchitis and emphysema (Petrache & Serban, 2023). Inflammation and excessive mucus secretion are common characteristics of chronic bronchitis (Kim & Criner, 2013; Widysanto & Mathew, 2022). In emphysema, the air sacs or alveoli in the lungs become enlarged and impaired, destroying lung tissue, decreasing airflow, and reducing lung function (Mouronte-Roibás et al., 2016; Petrache & Serban, 2023). Also, lung cancer risk increases in individuals exposed to CS (Abelia et al., 2023; Walser et al., 2008).

Understanding how cigarette smoking affects different parts of the respiratory system is crucial for improving clinical intervention and management of CS-related respiratory disorders. Since the lungs are vulnerable and have a high prevalence of diseases, they are often the focus

of CS inhalation research. However, to maintain respiratory health, it is essential to understand the concept of the unified airway theory (Bachert et al., 2023; Yii et al., 2018) which emphasizes the interconnected nature of the respiratory tract from the nose to the lungs. Therefore, an issue with the upper airways may affect the lower airways and vice versa. Thus, an integrated approach to respiratory care requires acknowledging this interconnection.

As a research associate at Stanford University in California, USA, I am investigating alterations in laryngeal transcriptome following CS inhalation in both *in vivo* animal models (Easwaran et al., 2021, 2022; Erickson-DiRenzo et al., 2020, 2021) and *in vitro* cell cultures (Martinez et al., 2023). The larynx, also known as the voice box, is a part of the respiratory system where the upper and lower airways meet. However, this organ has not been extensively studied. It is lined by two types of epithelia: stratified squamous epithelium in the vocal fold region and respiratory epithelium or pseudostratified columnar ciliated epithelium in the rest of the larynx (Easwaran et al., 2021). The respiratory tract is almost entirely covered by this respiratory epithelium as well (Kia'i & Bajaj, 2023). At present, the effects of CS on the laryngeal mucosal regions are not well established. Since most research regarding the impact of CS has been focused on the lower airways, lungs in particular, we generally rely on lower airway-related literature to make inferences. According to the unified airway theory (Bachert et al., 2023; Yii et al., 2018), responses in one airway region can affect responses in another. This implies that laryngeal epithelial cells may behave similarly to other types of epithelial cells found in other airway regions after CS exposures and vice versa.

Historically, *in vivo* models have been used to examine and gain a deeper understanding of such cellular responses to CS exposure. However, these models present technical, ethical, and financial challenges and may not fully capture the characteristics of the human airway microenvironment. In this regard, air-liquid interface (ALI) (Lacroix et al., 2018) is an *in vitro* technique that offers a cost-effective way to simulate the airway microenvironment. Since CS affects different types of airway epithelial cells differently, it is essential to study various cell types in both the basal and differentiated compartments of the epithelium. ALI cultures mimic the physiological conditions of the airways and provide an advantage to researchers in uncovering the complex molecular mechanisms that underlie exposure to CS in different cellular states. Several studies have used ALI models to analyze airway epithelial cells derived from human smokers (Abo et al., 2022; Benam et al., 2016; Gindele et al., 2020; Goldfarbmuren et al., 2020; Mathis et al., 2013; Moses et al., 2017; Purkayastha et al., 2020; Rager et al., 2013; Schamberger et al., 2015). With the motivation from my professional work and the knowledge gained in this course, I thus planned to extend my research beyond the larynx. I aim to investigate how CS affects non-laryngeal airway epithelial cells *in vitro* using ALI through this class project. I can draw inferences and better understand the subject by comparing these findings with my ongoing professional research on laryngeal transcriptomic changes in response to CS inhalation challenges.

Aims and Rationale

The purpose of this class project is to address the following aims:

- **Specific Aim 1.** Characterize the distinct transcriptomic responses of human and mouse respiratory epithelial cells to CS exposure in the non-laryngeal airway regions within ALI cell cultures.
- **Specific Aim 2.** Conduct a comparative analysis of the identified transcriptomic responses in human and mouse models to identify similarities and differences.
- **Specific Aim 3.** Contrast the transcriptomic outcomes of human and mouse non-laryngeal airway regions with their laryngeal counterparts, drawing insights from existing literature.

The rationale for this study is for me to delve into the transcriptomic responses of human and mouse respiratory epithelial cells when exposed to CS. By identifying common and distinct pathways activated in these cells, I aim to elucidate the conserved mechanisms and species-specific variations in response to CS. Additionally, I will compare the transcriptomic outcomes between non-laryngeal and laryngeal airway regions within each species to gain insight into any potential differences based on anatomical location. This comparative approach will enable me to understand CS-induced cellular responses better and help bridge the translational gap between preclinical models and human disease, thereby offering valuable insights for developing potential therapeutics.

Methods

Data retrieval

The human and mouse datasets used in this project were identified using the National Center for Biotechnology Information Gene Expression Omnibus (NCBI GEO) database (Barrett et al., 2013). The search query used was “cigarette smoke and air-liquid interface”. The identified datasets will be processed using the NIH LINCS tool called “GREIN” or GEO RNA-seq Experiments Interactive Navigator to acquire raw data counts for human and mouse datasets (Mahi et al., 2019). The final analysis used two Bulk RNA-seq studies, one addressing human bronchial epithelial cells (GSE152446) and one addressing mice tracheal epithelial cells (GSE229038).

Exploratory data analysis

NCBI GEO-retrieved datasets were initially subjected to exploratory data analysis (EDA), such as Principal Component Analysis (PCA), to investigate the inherent structure and relationships among all study samples. The “prcomp” function from the R programming environment “stats” package (version 3.6.2) was used for this purpose. A hierarchical clustering of the datasets was also performed on these datasets using the “ComplexHeatmap” package in R (Gu et al., 2016).

Differential expression analysis

Differential expression analysis was carried out utilizing the “DESeq2” R package (Love et al., 2014). The primary comparison was made between individuals who smoke and those who do not in the human dataset, as well as male mice exposed to CS and mice exposed to air as controls in the mouse dataset. The datasets were filtered based on $\text{abs}(\text{fold changes}) > 1.5$ (equivalent to $> \sim 0.6$ in $\log_2$ space) and $p\text{-values} (\leq 0.05)$ to identify significant differentially expressed genes (DEGs). The $p\text{-values}$ considered here were the Benjamini-Hochberg method-adjusted $p\text{-values}$ (Jafari & Ansari-Pour, 2019). The DEGs that passed the defined statistical significance levels were to be displayed as volcano plots using the “EnhancedVolcano” R package (*Bioconductor - EnhancedVolcano*, n.d.). Prior to this analysis, it is to be noted that the rows with low gene counts (< 10) were filtered.

Functional enrichment analysis

The significant DEGs underwent functional enrichment analysis employing Gene Ontology (GO) (Ashburner et al., 2000) and pathway databases such as Kyoto Encyclopedia of Genes and Genomes (KEGG) (Kanehisa, 2019; Kanehisa et al., 2023; Kanehisa & Goto, 2000), Reactome (Griss et al., 2020; Milacic et al., 2024), and WikiPathways (Agrawal et al., 2024; Pico et al., 2008) through Metascape (Zhou et al., 2019). The level of significance for each enriched term was considered as minimum gene overlap: 3, $p\text{-value cutoff}$: 0.05, and minimum enrichment: 1.5. The $p\text{-values}$ from Metascape enrichments are Benjamini-Hochberg method adjusted $p\text{-values}$. Additionally, the Molecular Complex Detection (MCODE) algorithm (Bader & Hogue, 2003)

integrated within Metascape was to be used to assist in identifying densely clustered modules of gene networks and their roles in biological functions.

Results

Positive results

Data retrieval

After conducting a preliminary search on the NCBI GEO database, 13 unique studies were found (**Table 1**). These studies used various techniques such as microarrays, bulk, and single-cell RNA-sequencing (Bulk RNA-seq and scRNA-seq). Most of the studies focused on humans (**Table 1**) and primarily evaluated respiratory epithelial cells from different regions such as nasal, tracheal, bronchial, alveolar type II, and tracheal regions. Out of these studies, eight studies provided well-documented evidence of CS-induced transcriptomic alterations in ALI cultures of different regions (Abo et al., 2022; Benam et al., 2016; Ghosh et al., 2023; Gindele et al., 2020; Goldfarbmuren et al., 2020; Moses et al., 2017; Purkayastha et al., 2020; Rager et al., 2013). The final analysis used two Bulk RNA-seq studies, one addressing human bronchial epithelial cells (GSE152446) and one addressing mice tracheal epithelial cells (GSE229038).

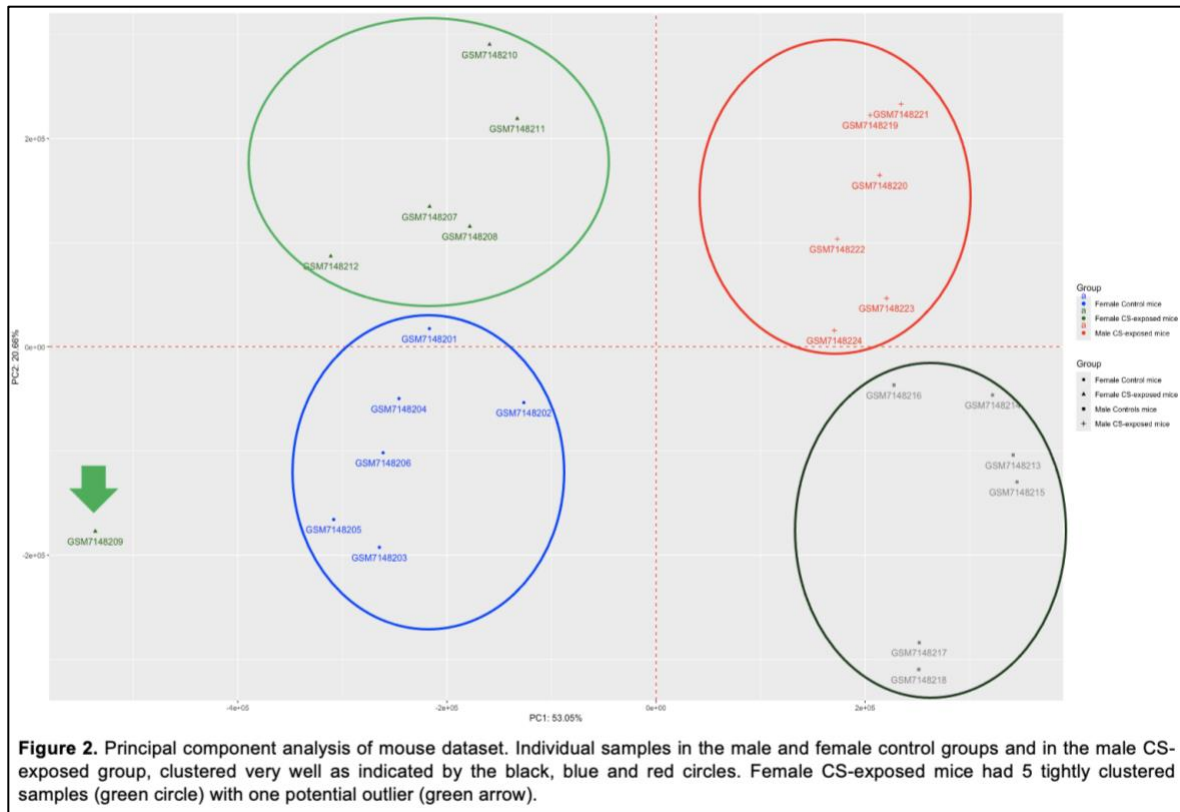
Table 1. GEO search results with cigarette smoke and air-liquid interface

Organism	Assay type	Airway Type	Specific Technique	GEO Accession ID
Homo sapiens	Expression profiling by array	Bronchial epithelial cells	scRNA-seq	GSE208662
Homo sapiens	Expression profiling by high throughput sequencing	Induced human pluripotent stem cells (iPSCs)-derived type 2 alveolar epithelial cells	Bulk RNA-seq	GSE223077
Homo sapiens	Expression profiling by high throughput sequencing	Pluripotent stem cell-derived alveolar type 2 cells (iAT2s)	Bulk RNA-seq (GSE184143) and scRNA-seq (GSE184142)	GSE184144 (super series)
Homo sapiens	Expression profiling by high throughput sequencing	Bronchial epithelial cells	scRNA-seq	GSE161089
Homo sapiens	Expression profiling by high throughput sequencing	Bronchial epithelial cells	Bulk RNA-seq	GSE152446
Homo sapiens	Expression profiling by high throughput sequencing	Airway cells	Bulk RNA-seq	GSE142100
Homo sapiens	Expression profiling by high throughput sequencing	Tracheal epithelial cells	scRNA-seq	GSE134174
Homo sapiens	Expression profiling by high throughput sequencing	Small airway epithelial cells (SAEC)	Bulk RNA-seq	GSE135188

Homo sapiens	Expression profiling by array	Human bronchial epithelial cells (HBECS)	Microarray analysis	GSE82137
Homo sapiens	Expression profiling by array	Human bronchiolar epithelium	Affymetrix microarray analysis	GSE87098
Homo sapiens	Methylation profiling by array	Nasal epithelial cells	DNA methylation array	GSE28368
Homo sapiens	Expression profiling by array	Lung airway epithelial cultures	Affymetrix microarray analysis	GSE30660
Mus musculus	Expression profiling by high throughput sequencing	Tracheal epithelial cells	Bulk RNA-seq	GSE229038

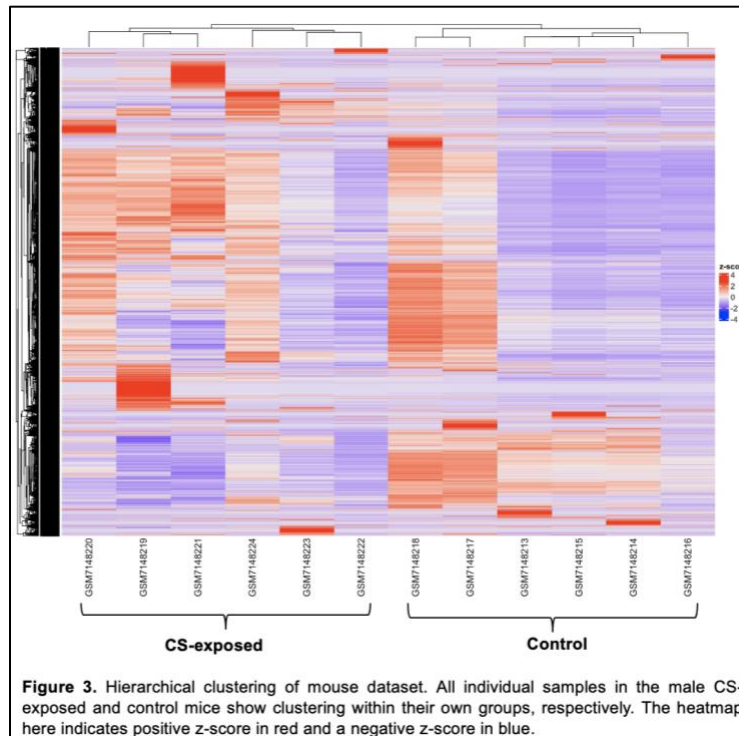
Mouse dataset EDA

PCA plot of the mouse tracheal dataset indicated the presence of 24 samples in total (**Figure 2**). The grouping of six male control mice and six male CS-exposed mice showed robust clustering, with no outlier samples present. Also, six control female mice displayed a tight clustering. However, an outlier was detected in the CS-exposed female mice. Overall, both male control and CS-exposed mice and their female counterparts displayed dispersion on both sides of the plot, in alignment with the red vertical axis line of symmetry. This suggested the presence of distinct gene expression patterns among these groups. For this class project, six samples from the male CS-exposed group and six from the male control group were filtered for subsequent analysis. The male mice were considered in order to be consistent with the male mice models that I use for my professional research (Easwaran et al., 2021, 2022; Erickson-DiRenzo et al., 2021) and not to introduce gender-based differences.



Hierarchical clustering of mouse dataset

In the global gene expression analysis context, it was observed that the individual male control mice samples and CS-exposed mice samples displayed distinct clustering patterns. This was identified by the proximity they shared with their respective counterparts within two distinct clusters, as indicated by the bifurcating dendrogram on the heat map (**Figure 3**). These clustering patterns indicated the possible existence of unique gene expression profiles for each group.



Differential expression analysis of mouse dataset

As indicated by the red dots in **Figure 4**, 2806 significant DEGs were identified post-analysis. Among these 2806 genes, 1229 were upregulated, and 1577 were downregulated.

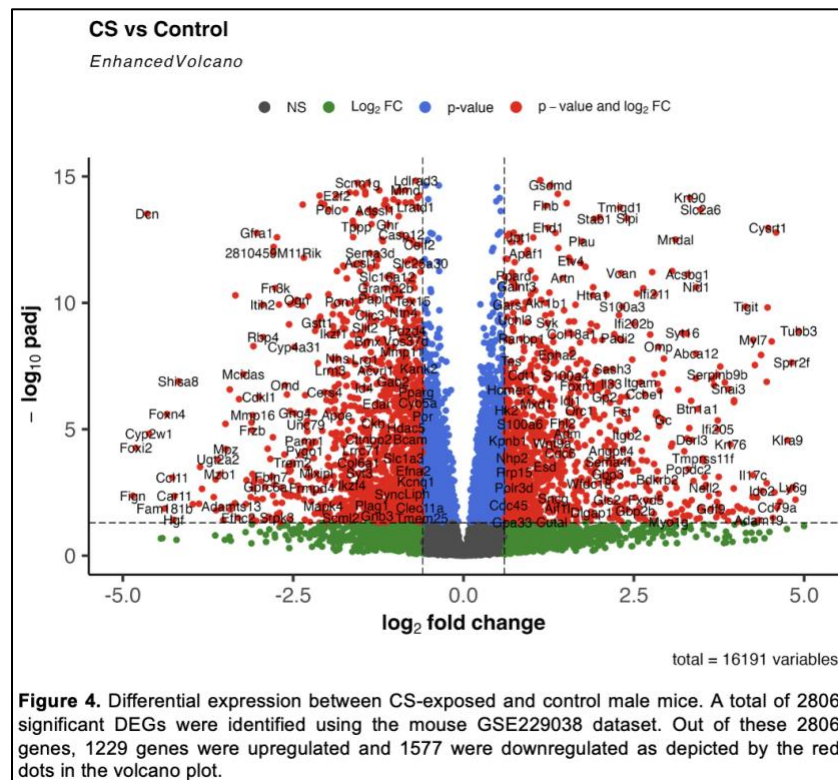


Figure 4. Differential expression between CS-exposed and control male mice. A total of 2806 significant DEGs were identified using the mouse GSE229038 dataset. Out of these 2806 genes, 1229 genes were upregulated and 1577 were downregulated as depicted by the red dots in the volcano plot.

Metascape functional enrichment analysis

Various pathways or functions were enriched via Metascape enrichment analysis, including extracellular matrix, biological oxidations, oxidoreductase activity, cancer pathways, and epithelial barrier-related pathways (basal cell adhesion and cell-cell adhesion), as shown in **Figure 5**. In fact, some of these enrichments (**Figure 6**) were similar to the previous work I conducted on the larynx (Easwaran et al., 2021, 2022; Erickson-DiRenzo et al., 2021).

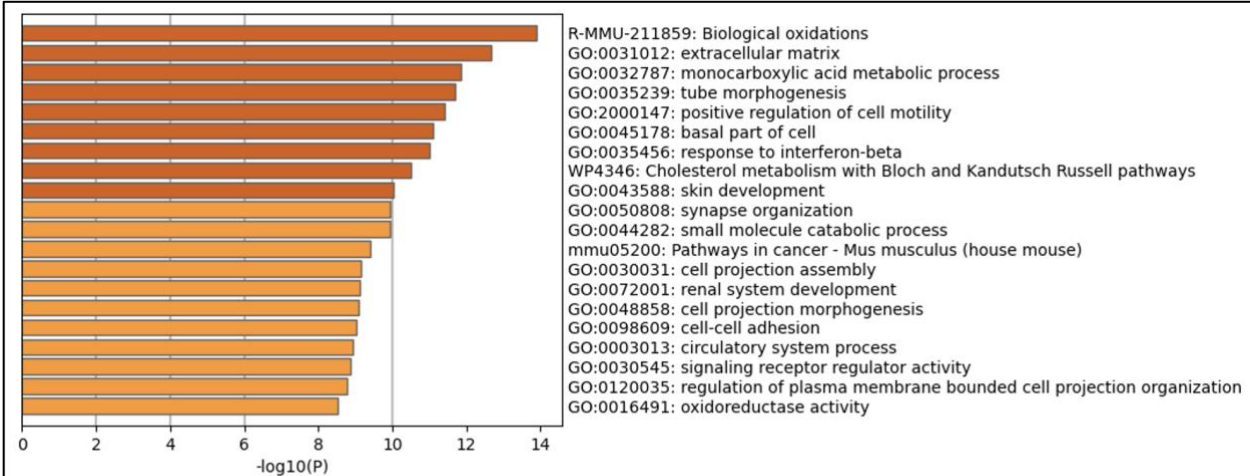


Figure 5. Gene ontology and Pathway database enrichments on Metascape. Significant DEGs were subjected to functional enrichment analysis utilizing Gene Ontology (GO) and pathway databases (Kyoto Encyclopedia of Genes and Genomes: KEGG, Reactome, and WikiPathways) with Metascape. The outcomes revealed enrichment in multiple biological functions such as extracellular matrix, biological oxidations or oxidoreductase activity, cancer pathways, and epithelial barrier related (basal cell and cell-cell adhesion).

GO	Category	Description	Count	%	Log10(P)	Log10(q)
R-MMU-211859	Reactome Gene Sets	Biological oxidations	66	2.56	-13.91	-9.52
GO:0031012	GO Cellular Components	extracellular matrix	123	4.76	-12.67	-8.64
GO:0032787	GO Biological Processes	monocarboxylic acid metabolic process	130	5.03	-11.85	-8.06
GO:0035239	GO Biological Processes	tube morphogenesis	160	6.19	-11.69	-8.00
GO:2000147	GO Biological Processes	positive regulation of cell motility	139	5.38	-11.43	-7.82
GO:0045178	GO Cellular Components	basal part of cell	86	3.33	-11.12	-7.63
GO:0035456	GO Biological Processes	response to interferon-beta	32	1.24	-11.01	-7.62
WP4346	WikiPathways	Cholesterol metabolism with Bloch and Kandutsch Russell pathways	26	1.01	-10.50	-7.22
GO:0043588	GO Biological Processes	skin development	79	3.06	-10.03	-6.81
GO:0050808	GO Biological Processes	synapse organization	87	3.37	-9.96	-6.81
GO:0044282	GO Biological Processes	small molecule catabolic process	82	3.17	-9.94	-6.81
mmu05200	KEGG Pathway	Pathways in cancer - Mus musculus (house mouse)	117	4.53	-9.40	-6.34
GO:0030031	GO Biological Processes	cell projection assembly	101	3.91	-9.14	-6.17
GO:0072001	GO Biological Processes	renal system development	82	3.17	-9.13	-6.17
GO:0048858	GO Biological Processes	cell projection morphogenesis	118	4.57	-9.11	-6.17
GO:0098609	GO Biological Processes	cell-cell adhesion	102	3.95	-9.03	-6.12
GO:0003013	GO Biological Processes	circulatory system process	106	4.10	-8.93	-6.05
GO:0030545	GO Molecular Functions	signaling receptor regulator activity	124	4.80	-8.88	-6.03
GO:0120035	GO Biological Processes	regulation of plasma membrane bounded cell projection organization	153	5.92	-8.77	-5.94
GO:0016491	GO Molecular Functions	oxidoreductase activity	152	5.88	-8.52	-5.75

Figure 6. Gene ontology and Pathway database enrichments on Metascape in tabular format. Various pathways or functions such as extracellular matrix, oxidative stress, cancer pathways, and epithelial barrier related like basal cell and cell-cell adhesion stood out as overlapping enrichments with those previously found in the larynx as indicated by the red rectangular boxes.

Negative results

Exclusion of human dataset based on EDA

The PCA plot showed that the human bronchial dataset had 16 samples (**Figure 7**). There was no tight grouping among the two male non-smokers, and the clustering appeared relatively dispersed. However, within the male smoker subset, one out of six subjects were outside the central cluster, which suggested a potential outlier. For the female subset, six non-smokers were identified, displaying a scattering distribution without any noticeable clustering. In contrast, the female smoker category had only two samples, one closer to the male smoker group, deviating from the expected clustering patterns. Overall, observations across gender categories indicated that male smokers and non-smokers, as well as their female counterparts, were dispersed across the PCA plot, aligning along the horizontal red axis line of symmetry on the top and bottom sides of the plot, respectively.

Initially, the plan was to extract information only from male human smokers and non-smokers for consistent gender matching with the mice dataset utilized in this class project and my professional research work (Easwaran et al., 2021, 2022; Erickson-DiRenzo et al., 2021). However, upon review of the sample sizes, particularly within the male non-smoker group (**Table 2**), it became evident that this study might not provide adequate data for robust statistical analysis. Consequently, all subsequent analyses omitted the human dataset and focused solely on the mouse datasets. This decision had implications for the original three specific aims, which centered on comparing the two datasets and identifying commonalities and differences in the response of non-laryngeal airway regions to CS exposure in both human and mouse species.

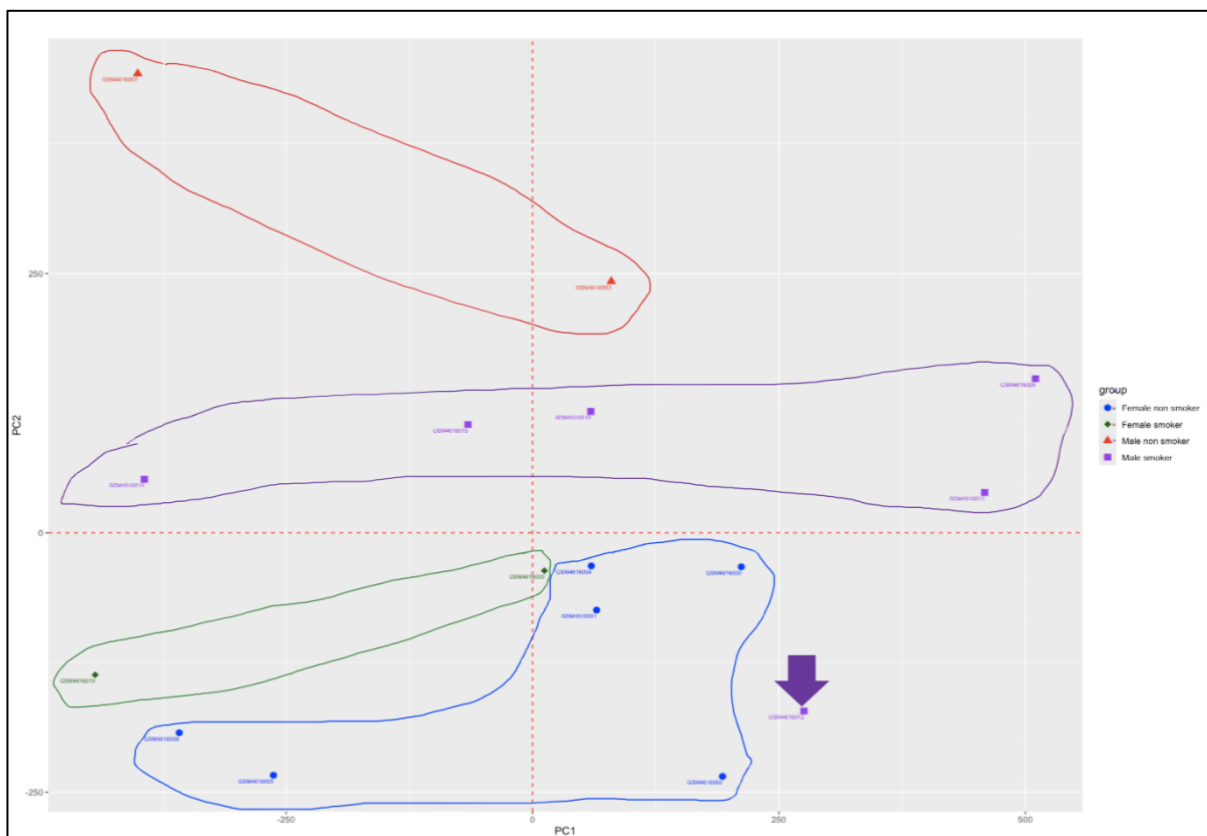


Figure 7. Principal component analysis of human dataset. Individual samples in the male and female non-smoker groups was dispersed and not very tightly clustered as indicated red and blue colored shapes and tracings, respectively. Male smoker group had one potential outlier (purple arrow). In the female smoker group, there were only two samples (green diamond shape and tracing), out of which one was clustered closer to the male smoker group.

Table 2. GSE152446 group sample sizes

Samples	Characteristics	Sample size
GSM4616000	Female non-smoker	n = 6
GSM4616001	Female non-smoker	
GSM4616002	Female non-smoker	
GSM4616004	Female non-smoker	
GSM4616005	Female non-smoker	
GSM4616006	Female non-smoker	
GSM4616009	Female smoker	n = 2
GSM4616013	Female smoker	n = 2
GSM4616003	Male non-smoker	
GSM4616007	Male non-smoker	
GSM4616008	Male smoker	n = 6
GSM4616010	Male smoker	
GSM4616011	Male smoker	
GSM4616012	Male smoker	
GSM4616014	Male smoker	
GSM4616015	Male smoker	

MCODE Network analysis

The MCODE plugin within Metascape encountered difficulties processing the 2806 DEGs for the network analysis (**Figure 8**). The inability to apply MCODE for network analysis hampered the identification of densely connected regions or critical pathways within the gene expression data, which could have provided valuable insights into the underlying biological mechanisms complementary to the pathway database enrichments.

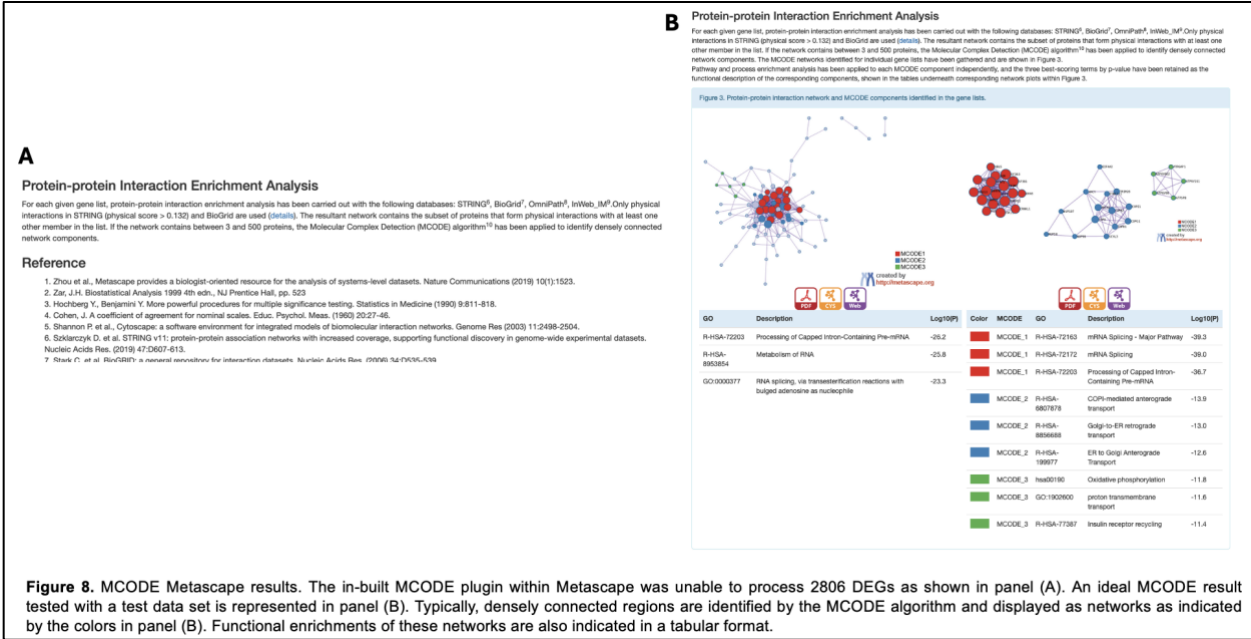


Figure 8. MCODE Metascape results. The in-built MCODE plugin within Metascape was unable to process 2806 DEGs as shown in panel (A). An ideal MCODE result tested with a test data set is represented in panel (B). Typically, densely connected regions are identified by the MCODE algorithm and displayed as networks as indicated by the colors in panel (B). Functional enrichments of these networks are also indicated in a tabular format.

Limitations and Alternative Approaches

This class project had limitations that required alternative approaches to be considered. One challenge was that using Bulk RNA-seq analysis made it difficult to capture individual cellular differences after CS exposures, especially for rare cell types. Focusing solely on mRNA or messenger RNA via Bulk RNA-seq datasets also ignored other RNA forms and epigenetic modifications influenced by CS exposures. Therefore, it would be beneficial to consider alternative methods, such as scRNA-seq and DNA methylation analysis, to uncover cellular heterogeneity and epigenetic information.

Bulk RNA-seq was also susceptible to technical noise and batch effects, which was particularly problematic when combining data from different studies or platforms such as NCBI GEO. Pre-defined statistical models and thresholds could affect differential expression and functional enrichment analyses, risking critical gene expression data loss or overestimating transcriptomic changes.

Additionally, the limited availability of Bulk RNA-seq and ALI-specific datasets across mice and humans posed a significant challenge to conducting comprehensive inter-species comparisons. To align with the original aims of the study, only one study specific to each species was selected. However, this approach encountered an additional obstacle as the chosen studies focused on epithelial cells from different airway regions, one examining the trachea and the other the bronchi. Ideally, comparing the same airway region in both species would have been preferable. Nonetheless, it is worth noting that the bronchi and trachea exhibit the same type of respiratory epithelium (Musah et al., 2012).

Consequently, another major shift in approach was necessary. Insufficient sample sizes, particularly evident in the human male cohorts within the chosen dataset, compromised statistical power and result validity, necessitating their exclusion from the study. To address such challenges effectively, conducting power analysis and ensuring sufficient sample sizes per group is recommended.

Furthermore, excluding female cohorts overlooked potential sex-specific differences in CS-induced responses, reducing the comprehensiveness and applicability of findings. Lastly, incompatibility issues with network analysis tools like MCODE prompted the consideration of alternative methods such as StringDB (Szklarczyk et al., 2021) or Weighted Gene Co-expression Network Analysis (WGCNA) (Langfelder & Horvath, 2008) for future research.

Despite these challenges, the findings of this class project establish the existence of analogous characteristics between CS-exposed mouse laryngeal and tracheal epithelial cells under ALI conditions. Future directions warrant further investigation into inter-species comparisons post-CS exposures in ALI cultures.

References

- Abelia, X. A., Lesmana, R., Goenawan, H., Abdulah, R., & Barliana, M. I. (2023). Comparison impact of cigarettes and e-cigs as lung cancer risk inductor: a narrative review. *European Review for Medical and Pharmacological Sciences*, 27(13), 6301–6318. https://doi.org/10.26355/EURREV_202307_32990
- Abo, K. M., de Aja, J. S., Lindstrom-Vautrin, J., Alysandratos, K. D., Richards, A., Garcia-De-Alba, C., Huang, J., Hix, O. T., Werder, R. B., Bullitt, E., Hinds, A., Falconer, I., Villacorta-Martin, C., Jaenisch, R., Kim, C. F., Kotton, D. N., & Wilson, A. A. (2022). Air-liquid interface culture promotes maturation and allows environmental exposure of pluripotent

- stem cell–derived alveolar epithelium. *JCI Insight*, 7(6).
<https://doi.org/10.1172/JCI.INSIGHT.155589>
- Agrawal, A., Balci, H., Hanspers, K., Coort, S. L., Mar ens, M., Slenter, D. N., ieder ike Ehr hart, F., Digles, D., Waagmeest er, A., Wassink, I., Abbassi-Daloi, T., Lopes, E. N., ary ly er, A., vier Millán Acosta, J., Willighag en, L. G., Nishida, K., Riutta, A., Basar ic, H., is Evelo, C. T., ... Pico, A. R. (2024). WikiPathways 2024: next generation pathway database. *Nucleic Acids Research*, 52(D1), D679–D689. <https://doi.org/10.1093/NAR/GKAD960>
- Ashburner, M., Ball, C. A., Blake, J. A., Botstein, D., Butler, H., Cherry, J. M., Davis, A. P., Dolinski, K., Dwight, S. S., Eppig, J. T., Harris, M. A., Hill, D. P., Issel-Tarver, L., Kasarskis, A., Lewis, S., Matese, J. C., Richardson, J. E., Ringwald, M., Rubin, G. M., & Sherlock, G. (2000). Gene Ontology: tool for the unification of biology. *Nature Genetics* 25:1, 25(1), 25–29. <https://doi.org/10.1038/75556>
- Bachert, C., Luong, A. U., Gevaert, P., Mullol, J., Smith, S. G., Silver, J., Sousa, A. R., Howarth, P. H., Benson, V. S., Mayer, B., Chan, R. H., & Busse, W. W. (2023). The Unified Airway Hypothesis: Evidence From Specific Intervention With Anti–IL-5 Biologic Therapy. *The Journal of Allergy and Clinical Immunology: In Practice*, 11(9), 2630–2641.
<https://doi.org/10.1016/J.JAIP.2023.05.011>
- Bader, G. D., & Hogue, C. W. V. (2003). An automated method for finding molecular complexes in large protein interaction networks. *BMC Bioinformatics*, 4(1), 1–27.
<https://doi.org/10.1186/1471-2105-4-2/FIGURES/12>
- Barrett, T., Wilhite, S. E., Ledoux, P., Evangelista, C., Kim, I. F., Tomashevsky, M., Marshall, K. A., Phillippy, K. H., Sherman, P. M., Holko, M., Yefanov, A., Lee, H., Zhang, N., Robertson, C. L., Serova, N., Davis, S., & Soboleva, A. (2013). NCBI GEO: archive for functional genomics data sets—update. *Nucleic Acids Research*, 41(D1), D991–D995.
<https://doi.org/10.1093/NAR/GKS1193>
- Benam, K. H., Novak, R., Nawroth, J., Hirano-Kobayashi, M., Ferrante, T. C., Choe, Y., Prantil-Baun, R., Weaver, J. C., Bahinski, A., Parker, K. K., & Ingber, D. E. (2016). Matched-Comparative Modeling of Normal and Diseased Human Airway Responses Using a Microengineered Breathing Lung Chip. *Cell Systems*, 3(5), 456-466.e4.
<https://doi.org/10.1016/j.cels.2016.10.003>
- Bioconductor - EnhancedVolcano. (n.d.). Retrieved April 24, 2024, from
<https://bioconductor.org/packages/release/bioc/html/EnhancedVolcano.html>
- Centers for Disease Control and Prevention, N. C. for C. D. P. and H. P. O. on S. and H. (2010). How Tobacco Smoke Causes Disease: The Biology and Behavioral Basis for Smoking-Attributable Disease: A Report of the Surgeon General. Atlanta (GA): Centers for Disease Control and Prevention (US). In *How Tobacco Smoke Causes Disease: The Biology and Behavioral Basis for Smoking-Attributable Disease: A Report of the Surgeon General*. Centers for Disease Control and Prevention (US).
<https://www.ncbi.nlm.nih.gov/books/NBK53017/>
- Curtin, G. M., Higuchi, M. A., Ayres, P. H., Swauger, J. E., & Mosberg, A. T. (2004). Lung tumorigenicity in A/J and rasH2 transgenic mice following mainstream tobacco smoke inhalation. *Toxicological Sciences*, 81(1), 26–34. <https://doi.org/10.1093/toxsci/kfh175>
- Easwaran, M., Martinez, J. D., Kim, J. B., & Erickson-DiRenzo, E. (2022). Modulation of mouse laryngeal inflammatory and immune cell responses by low and high doses of mainstream cigarette smoke. *Scientific Reports* 2022 12:1, 12(1), 1–19. <https://doi.org/10.1038/s41598-022-23359-7>
- Easwaran, M., Martinez, J. D., Ramirez, D. J., Gall, P. A., & Erickson-DiRenzo, E. (2021). Short-term whole body cigarette smoke exposure induces regional differences in cellular response in the mouse larynx. *Toxicology Reports*, 8, 920–937.
<https://doi.org/10.1016/J.TOXREP.2021.04.007>

- Erickson-DiRenzo, E., Easwaran, M., Martinez, J. D., Dewan, K., & Sung, C. K. (2021). Mainstream Cigarette Smoke Impacts the Mouse Vocal Fold Epithelium and Mucus Barrier. *The Laryngoscope*, 131(11), 2530–2539. <https://doi.org/10.1002/LARY.29572>
- Erickson-DiRenzo, E., Singh, S. P., Martinez, J. D., Sanchez, S. E., Easwaran, M., & Valdez, T. A. (2020). Cigarette smoke-induced changes in the murine vocal folds: A Raman spectroscopic observation. *Analyst*, 145(23), 7709–7717. <https://doi.org/10.1039/d0an01570a>
- Garcia Martins, R. H., Marques Madeira, S. L., Fabro, A. T., Rocha, N. D. S., De Oliveira Semenzati, G., & Alves, K. F. (2012). Effects to exposure of tobacco smoke and alcohol on the tongue and pharynx of rats. *Inhalation Toxicology*, 24(3), 153–160. <https://doi.org/10.3109/08958378.2011.649190>
- Ghosh, B., Chengala, P. P., Shah, S., Chen, D., Karnam, V., Wilmsen, K., Yeung-Luk, B., & Sidhaye, V. K. (2023). Cigarette smoke-induced injury induces distinct sex-specific transcriptional signatures in mice tracheal epithelial cells. *American Journal of Physiology. Lung Cellular and Molecular Physiology*, 325(4), L467–L476. <https://doi.org/10.1152/AJPLUNG.00104.2023/ASSET/IMAGES/MEDIUM/L-00104-2023R01.PNG>
- Gindele, J. A., Kiechle, T., Benediktus, K., Birk, G., Brendel, M., Heinemann, F., Wohnhaas, C. T., LeBlanc, M., Zhang, H., Strulovici-Barel, Y., Crystal, R. G., Thomas, M. J., Stierstorfer, B., Quast, K., & Schymeinsky, J. (2020). Intermittent exposure to whole cigarette smoke alters the differentiation of primary small airway epithelial cells in the air-liquid interface culture. *Scientific Reports* 2020 10:1, 10(1), 1–17. <https://doi.org/10.1038/s41598-020-63345-5>
- Goldfarbmuren, K. C., Jackson, N. D., Sajuthi, S. P., Dyjack, N., Li, K. S., Rios, C. L., Plender, E. G., Montgomery, M. T., Everman, J. L., Bratcher, P. E., Vladar, E. K., & Seibold, M. A. (2020). Dissecting the cellular specificity of smoking effects and reconstructing lineages in the human airway epithelium. *Nature Communications* 2020 11:1, 11(1), 1–21. <https://doi.org/10.1038/s41467-020-16239-z>
- Griss, J., Viteri, G., Sidiropoulos, K., Nguyen, V., Fabregat, A., & Hermjakob, H. (2020). ReactomeGSA - Efficient Multi-Omics Comparative Pathway Analysis. *Molecular & Cellular Proteomics : MCP*, 19(12), 2115–2124. <https://doi.org/10.1074/MCP.TIR120.002155>
- Gu, Z., Eils, R., & Schlesner, M. (2016). Complex heatmaps reveal patterns and correlations in multidimensional genomic data. *Bioinformatics*, 32(18), 2847–2849. <https://doi.org/10.1093/BIOINFORMATICS/BTW313>
- Hadar, T., Yaniv, E., Shvili, Y., Koren, R., & Shvero, J. (2009). Histopathological changes of the nasal mucosa induced by smoking Smoking-induced nasal mucosa changes. *Inhalation Toxicology*, 21(13), 1119–1122. <https://doi.org/10.3109/08958370902767070>
- Jafari, M., & Ansari-Pour, N. (2019). Why, When and How to Adjust Your P Values? *Cell Journal (Yakhteh)*, 20(4), 604. <https://doi.org/10.22074/CELLJ.2019.5992>
- Kanehisa, M. (2019). Toward understanding the origin and evolution of cellular organisms. *Protein Science*, 28(11), 1947–1951. <https://doi.org/10.1002/PRO.3715>
- Kanehisa, M., Furumichi, M., Sato, Y., Kawashima, M., & Ishiguro-Watanabe, M. (2023). KEGG for taxonomy-based analysis of pathways and genomes. *Nucleic Acids Research*, 51(D1), D587–D592. <https://doi.org/10.1093/NAR/GKAC963>
- Kanehisa, M., & Goto, S. (2000). KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, 28(1), 27–30. <https://doi.org/10.1093/NAR/28.1.27>
- Kia'i, N., & Bajaj, T. (2023). Histology, Respiratory Epithelium. *StatPearls*. <https://www.ncbi.nlm.nih.gov/books/NBK541061/>
- Kim, V., & Criner, G. J. (2013). Chronic Bronchitis and Chronic Obstructive Pulmonary Disease. *American Journal of Respiratory and Critical Care Medicine*, 187(3), 228. <https://doi.org/10.1164/RCCM.201210-1843CI>

- Lacroix, G., Koch, W., Ritter, D., Gutleb, A. C., Larsen, S. T., Loret, T., Zanetti, F., Constant, S., Chortarea, S., Rothen-Rutishauser, B., Hiemstra, P. S., Frejafon, E., Hubert, P., Gribaldo, L., Kearns, P., Aublant, J. M., Diabaté, S., Weiss, C., De Groot, A., & Kooter, I. (2018). Air-Liquid Interface in Vitro Models for Respiratory Toxicology Research: Consensus Workshop and Recommendations. *Applied In Vitro Toxicology*, 4(2), 91–106. <https://doi.org/10.1089/AIVT.2017.0034/ASSET/IMAGES/LARGE/FIGURE4.JPEG>
- Langfelder, P., & Horvath, S. (2008). WGCNA: An R package for weighted correlation network analysis. *BMC Bioinformatics*, 9(1), 1–13. <https://doi.org/10.1186/1471-2105-9-559/FIGURES/4>
- Laniado-Laborin, R. (2009). Smoking and Chronic Obstructive Pulmonary Disease (COPD). Parallel Epidemics of the 21st Century. *International Journal of Environmental Research and Public Health*, 6(1), 209. <https://doi.org/10.3390/IJERPH6010209>
- Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15(12), 1–21. <https://doi.org/10.1186/S13059-014-0550-8/FIGURES/9>
- Mahi, N. Al, Najafabadi, M. F., Pilarczyk, M., Kouril, M., & Medvedovic, M. (2019). GREIN: An Interactive Web Platform for Re-analyzing GEO RNA-seq Data. *Scientific Reports*, 9(1). <https://doi.org/10.1038/S41598-019-43935-8>
- Martinez, J. D., Easwaran, M., Ramirez, D., & Erickson-DiRenzo, E. (2023). Effects of Electronic (E)-cigarette Vapor and Cigarette Smoke in Cultured Vocal Fold Fibroblasts. *The Laryngoscope*, 133(1), 139–146. <https://doi.org/10.1002/LARY.30073>
- Mathis, C., Poussin, C., Weisensee, D., Gebel, S., Hengstermann, A., Sewer, A., Belcastro, V., Xiang, Y., Ansari, S., Wagner, S., Hoeng, J., & Peitsch, M. C. (2013). Human bronchial epithelial cells exposed in vitro to cigarette smoke at the air-liquid interface resemble bronchial epithelium from human smokers. *American Journal of Physiology - Lung Cellular and Molecular Physiology*, 304(7), L489. <https://doi.org/10.1152/AJPLUNG.00181.2012>
- Milacic, M., Beavers, D., Conley, P., Gong, C., Gillespie, M., Griss, J., Haw, R., Jassal, B., Matthews, L., May, B., Petryszak, R., Ragueneau, E., Rothfels, K., Sevilla, C., Shamovsky, V., Stephan, R., Tiwari, K., Varusai, T., Weiser, J., ... D'Eustachio, P. (2024). The Reactome Pathway Knowledgebase 2024. *Nucleic Acids Research*, 52(D1), D672–D678. <https://doi.org/10.1093/NAR/GKAD1025>
- Moore, C. (1971). Cigarette Smoking and Cancer of the Mouth, Pharynx, and Larynx: A Continuing Study. *JAMA*, 218(4), 553–558. <https://doi.org/10.1001/JAMA.1971.03190170031005>
- Moses, E., Wang, T., Corbett, S., Jackson, G. R., Drizik, E., Perdomo, C., Perdomo, C., Kleerup, E., Brooks, D., O'Connor, G., Dubinett, S., Hayden, P., Lenburg, M. E., & Spira, A. (2017). Molecular Impact of Electronic Cigarette Aerosol Exposure in Human Bronchial Epithelium. *Toxicological Sciences*, 155(1), 248–257. <https://doi.org/10.1093/TOXSCI/KFW198>
- Mouronte-Roibás, C., Leiro-Fernández, V., Fernández-Villar, A., Botana-Rial, M., Ramos-Hernández, C., & Ruano-Ravina, A. (2016). COPD, emphysema and the onset of lung cancer. A systematic review. *Cancer Letters*, 382(2), 240–244. <https://doi.org/10.1016/J.CANLET.2016.09.002>
- Musah, S., Chen, J., & Hoyle, G. W. (2012). Repair of tracheal epithelium by basal cells after chlorine-induced injury. *Respiratory Research*, 13(1), 1–12. <https://doi.org/10.1186/1465-9921-13-107/FIGURES/12>
- Pagliuca, G., Rosato, C., Martellucci, S., De Vincentiis, M., Greco, A., Fusconi, M., De Virgilio, A., Gallipoli, C., Simonelli, M., & Gallo, A. (2015). Cytologic and Functional Alterations of Nasal Mucosa in Smokers: Temporary or Permanent Damage? *Otolaryngology– Head and Neck Surgery*, 152(4), 740–745. <https://doi.org/10.1177/0194599814566598>

- Petrache, I., & Serban, K. (2023). Emphysema. *Pathobiology of Human Disease: A Dynamic Encyclopedia of Disease Mechanisms*, 2609–2624. <https://doi.org/10.1016/B978-0-12-386456-7.05305-3>
- Phillips, B., Veljkovic, E., Peck, M. J., Buettner, A., Elamin, A., Guedj, E., Vuillaume, G., Ivanov, N. V., Martin, F., Boué, S., Schlage, W. K., Schneider, T., Titz, B., Talikka, M., Vanscheeuwijck, P., Hoeng, J., & Peitsch, M. C. (2015). A 7-month cigarette smoke inhalation study in C57BL/6 mice demonstrates reduced lung inflammation and emphysema following smoking cessation or aerosol exposure from a prototypic modified risk tobacco product. *Food and Chemical Toxicology*, 80, 328–345. <https://doi.org/10.1016/j.fct.2015.03.009>
- Pico, A. R., Kelder, T., Van Iersel, M. P., Hanspers, K., Conklin, B. R., & Evelo, C. (2008). WikiPathways: Pathway Editing for the People. *PLOS Biology*, 6(7), e184. <https://doi.org/10.1371/JOURNAL.PBIO.0060184>
- Purkayastha, A., Sen, C., Garcia, G., Langerman, J., Shia, D. W., Meneses, L. K., Vijayaraj, P., Durra, A., Koloff, C. R., Freund, D. R., Chi, J., Rickabaugh, T. M., Mulay, A., Konda, B., Sim, M. S., Stripp, B. R., Plath, K., Arumugaswami, V., & Gomperts, B. N. (2020). Direct Exposure to SARS-CoV-2 and Cigarette Smoke Increases Infection Severity and Alters the Stem Cell-Derived Airway Repair Response. *Cell Stem Cell*, 27(6), 869. <https://doi.org/10.1016/J.STEM.2020.11.010>
- Radsel, Z., & Kambc, V. (1978). The influence of cigarette smoke on the pharyngeal mucosa. *Acta Oto-Laryngologica*, 85(1–2), 128–134. <https://doi.org/10.3109/00016487809121433>
- Rager, J. E., Bauer, R. N., Müller, L. L., Smeester, L., Carson, J. L., Brighton, L. E., Fry, R. C., & Jaspers, I. (2013). DNA methylation in nasal epithelial cells from smokers: Identification of ULBP3-related effects. *American Journal of Physiology - Lung Cellular and Molecular Physiology*, 305(6), 432–438. https://doi.org/10.1152/AJPLUNG.00116.2013/SUPPL_FILE/TABLES1.PDF
- Reh, D. D., Higgins, T. S., & Smith, T. L. (2012). Impact of tobacco smoke on chronic rhinosinusitis: a review of the literature. *International Forum of Allergy & Rhinology*, 2(5), 362–369. <https://doi.org/10.1002/alr.21054>
- Renne, R. A., & Gideon, K. M. (2006). Types and Patterns of Response in the Larynx Following Inhalation. *Toxicologic Pathology*, 34(3), 281–285. <https://doi.org/10.1080/01926230600695631>
- Renne, R., Brix, A., Harkema, J., Herbert, R., Kittel, B., Lewis, D., March, T., Nagano, K., Pino, M., Rittinghausen, S., Rosenbruch, M., Tellier, P., & Wohrmann, T. (2009). Proliferative and Nonproliferative Lesions of the Rat and Mouse Respiratory Tract. *Toxicologic Pathology*, 37(7_suppl), 5S-73S. <https://doi.org/10.1177/0192623309353423>
- Schamberger, A. C., Staab-Weijnitz, C. A., Mise-Racek, N., & Eickelberg, O. (2015). Cigarette smoke alters primary human bronchial epithelial cell differentiation at the air-liquid interface. *Scientific Reports* 2015 5:1, 5(1), 1–9. <https://doi.org/10.1038/srep08163>
- Services, U. S. D. of H. and H., Prevention, C. for D. C. and, Promotion, N. C. for C. D. P. and H., & Health, O. on S. and. (2014). *The Health Consequences of Smoking—50 Years of Progress*. Centers for Disease Control and Prevention (US). <https://doi.org/NBK179276>
- Song, Q., Chen, P., & Liu, X. M. (2021). The role of cigarette smoke-induced pulmonary vascular endothelial cell apoptosis in COPD. *Respiratory Research* 2021 22:1, 22(1), 1–15. <https://doi.org/10.1186/S12931-021-01630-1>
- Szklarczyk, D., Gable, A. L., Nastou, K. C., Lyon, D., Kirsch, R., Pyysalo, S., Doncheva, N. T., Legeay, M., Fang, T., Bork, P., Jensen, L. J., & von Mering, C. (2021). The STRING database in 2021: customizable protein–protein networks, and functional characterization of user-uploaded gene/measurement sets. *Nucleic Acids Research*, 49(D1), D605. <https://doi.org/10.1093/NAR/GKAA1074>

- Vineis, P., Alavanja, M., Buffler, P., Fontham, E., Franceschi, S., Gao, Y. T., Gupta, P. C., Hackshaw, A., Matos, E., Samet, J., Sitas, F., Smith, J., Stayner, L., Straif, K., Thun, M. J., Wichmann, H. E., Wu, A. H., Zaridze, D., Peto, R., & Doll, R. (2004). Tobacco and Cancer: Recent Epidemiological Evidence. *JNCI: Journal of the National Cancer Institute*, 96(2), 99–106. <https://doi.org/10.1093/JNCI/DJH014>
- Walser, T., Cui, X., Yanagawa, J., Lee, J. M., Heinrich, E., Lee, G., Sharma, S., & Dubinett, S. M. (2008). Smoking and Lung Cancer: The Role of Inflammation. *Proceedings of the American Thoracic Society*, 5(8), 811. <https://doi.org/10.1513/PATS.200809-100TH>
- Widysanto, A., & Mathew, G. (2022). Chronic Bronchitis. In *Essence of Anesthesia Practice E-Book*. StatPearls Publishing. <https://doi.org/10.1016/B978-1-4377-1720-4.00045-5>
- Yii, A. C. A., Tay, T. R., Choo, X. N., Koh, M. S. Y., Tee, A. K. H., & Wang, D. Y. (2018). Precision medicine in united airways disease: A “treatable traits” approach. *Allergy*, 73(10), 1964–1978. <https://doi.org/10.1111/ALL.13496>
- Zhou, Y., Zhou, B., Pache, L., Chang, M., Khodabakhshi, A. H., Tanaseichuk, O., Benner, C., & Chanda, S. K. (2019). Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. *Nature Communications* 2019 10:1, 10(1), 1–10. <https://doi.org/10.1038/s41467-019-09234-6>